Task 4: File Transfer Using SFTP and VSFTPD

Objective

Configure SFTP in an instance so that it can receive files through port 22 to a specific directory using a specific user. The SFTP users should have restricted privileges.

Verify the SFTP configuration using an FTP client such as FileZilla or Cyberduck.

Install and configure VSFTPD in an instance so that it can receive files securely through port 21 to the configured directory using the configured user. The VSFTPD users should also have restricted privileges.

Verify the VSFTPD configuration using an FTP client.

Requirements

- AWS EC2 instance running RHEL
- SFTP using OpenSSH
- VSFTPD FTP server
- User account `zenhal`
- SFTP directory `/sftp/zenhal/upload`
- VSFTPD directory `/vsftpd/zen/uploads`
- FileZilla FTP client
- AWS Security Group configuration
- RHEL firewalld configuration
- SELinux enabled
- TLS certificate for secure FTP communication

Procedure

Step 1 – Check SSH Service

SFTP works through the SSH service. The SSH service was checked using:

```
sudo systemctl status sshd
```

The SSH service was confirmed to be active and running.

Step 2 – Set Password for the SFTP User

The existing user `zenhal` was used for SFTP access.

```
sudo passwd zenhal
```

A password was configured for the user.

Step 3 – Create the SFTP Directory

A separate directory was created for restricted SFTP file transfers:

```
sudo mkdir -p /sftp/zenhal/upload
```

The directory structure was:

```
/sftp
└── zenhal
    └── upload
```

The `upload` directory was configured as the location where the SFTP user can transfer files.

Step 4 – Configure Directory Ownership

The SFTP root directories were assigned to the root user, while the upload directory was assigned to `zenhal`.

```
sudo chown root:root /sftp
sudo chown root:root /sftp/zenhal
sudo chown zenhal:zenhal /sftp/zenhal/upload
```

This prevents the SFTP user from modifying the chroot directories while allowing file transfers inside the `upload` directory.

Step 5 – Configure Restricted SFTP

The SSH configuration file was edited:

```
sudo nano /etc/ssh/sshd_config
```

The following configuration was added:

```
Match User zenhal
    ForceCommand internal-sftp
    ChrootDirectory /sftp/zenhal
    PasswordAuthentication yes
    AllowTcpForwarding no
    X11Forwarding no
```

This configuration restricts the `zenhal` user to SFTP access and prevents normal SSH shell access.

```
Match User zenhal
    ForceCommand internal-sftp
    ChrootDirectory /sftp/zenhal
    PasswordAuthentication yes
    AllowTcpForwarding no
    X11Forwarding no
```

Step 6 – Test and Apply the SSH Configuration

The SSH configuration was tested using:

```
sudo sshd -t
```

No error was returned, confirming that the configuration syntax was correct.

The SSH service was then restarted:

```
sudo systemctl restart sshd
```

Step 7 – Verify SFTP Using FileZilla

FileZilla was configured with the following settings:

```
Protocol: SFTP - SSH File Transfer Protocol
Host: <EC2-Public-IP>
Port: 22
Logon Type: Normal
User: zenhal
Password: <configured password>
```

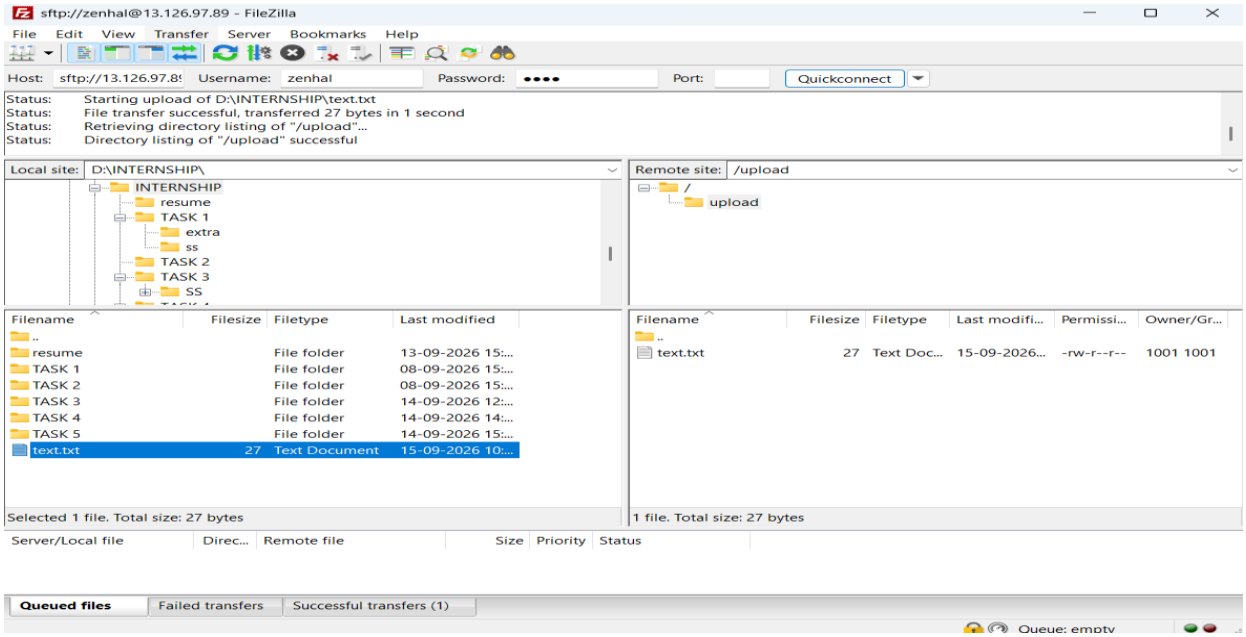
The connection was successfully established through port 22.

The restricted directory was displayed as:

```
/
└── upload
```

A test file was transferred through FileZilla to verify the SFTP configuration.

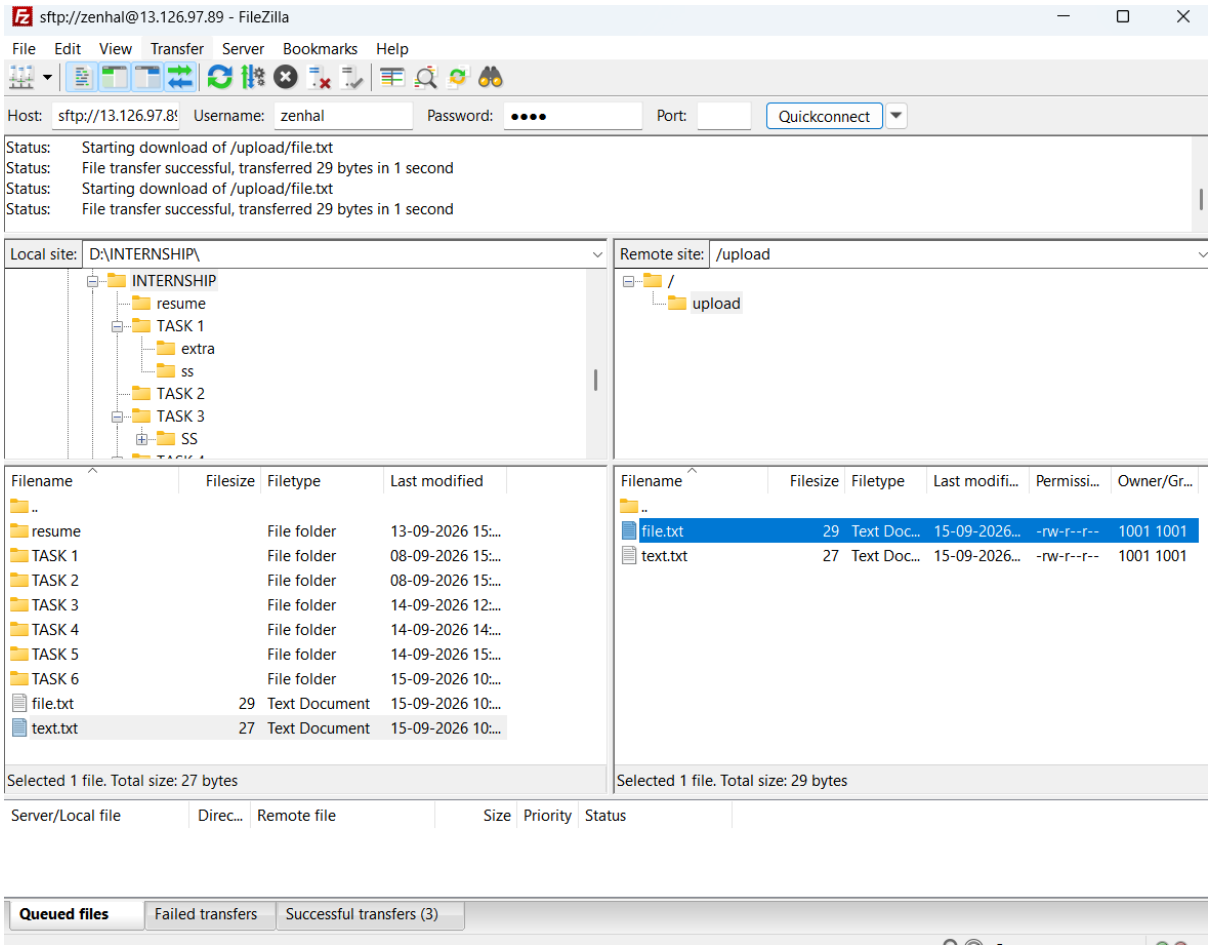
Windows to server:




```
[ec2-user@ip-172-31-7-163 ~]$ cd /sftp
[ec2-user@ip-172-31-7-163 sftp]$ cd zenhal
[ec2-user@ip-172-31-7-163 zenhal]$ cd upload
[ec2-user@ip-172-31-7-163 upload]$ ls
text.txt
[ec2-user@ip-172-31-7-163 upload]$ cat text.txt
hello this sftp sample file[ec2-user@ip-172-31-7-163 upload]$ ^C
```

Server to windows:

```
[ec2-user@ip-172-31-7-163 ~]$ su - zenhal
Password:
[zenhal@ip-172-31-7-163 ~]$ cd /sftp/zenhal/upload
[zenhal@ip-172-31-7-163 upload]$ cat >> file.txt
hello server to windows file
^C
[zenhal@ip-172-31-7-163 upload]$ cat file.txt
hello server to windows file
[zenhal@ip-172-31-7-163 upload]$ cd
```



VSFTPD

Step 8 – Install VSFTPD

The VSFTPD package was installed on the RHEL EC2 instance:

```
sudo dnf install vsftpd -y
```

VSFTPD provides FTP server functionality for file transfer.

Step 9 – Enable and Start VSFTPD

The VSFTPD service was enabled and started:

```
sudo systemctl enable --now vsftpd
```

The service status was checked using:

```
sudo systemctl status vsftpd
```

The VSFTPD service was confirmed to be active and running.

Step 10 – Create the VSFTPD Directory

A separate directory was created for VSFTPD file transfers:

```
sudo mkdir -p /vsftpd/zen/uploads
```

The directory structure was:

```
/vsftpd
└── zen
    └── uploads
```

Step 11 – Configure Directory Ownership

The main VSFTPD directories were assigned to root, while the upload directory was assigned to `zenhal`.

```
sudo chown root:root /vsftpd
```

```
sudo chown root:root /vsftpd/zen
sudo chown zenhal:zenhal /vsftpd/zen/uploads
```

This provides restricted access while allowing `zenhal` to upload files into the `uploads` directory.

Step 12 – Configure VSFTPD

The VSFTPD configuration file was edited:

```
sudo nano /etc/vsftpd/vsftpd.conf
```

The following settings were configured:

```
anonymous_enable=NO
local_enable=YES
write_enable=YES
local_umask=022
```

These settings disable anonymous access and allow authenticated local users to transfer files.

Step 13 – Restrict the FTP User

The following settings were added:

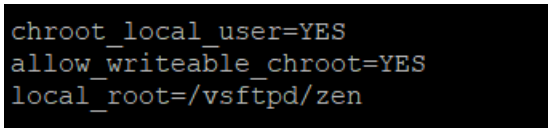
```
chroot_local_user=YES
allow_writeable_chroot=YES
local_root=/vsftpd/zen
```

This restricts the `zenhal` user to the configured VSFTPD directory.

The user can access:

```
/vsftpd/zen/uploads
```

but cannot freely browse other areas of the server through FTP.



Step 14 – Configure Passive FTP

Passive FTP was enabled using:

```
pasv_enable=YES
pasv_min_port=30000
pasv_max_port=30010
```

Port 21 is used for the FTP control connection, while ports 30000–30010 are used for passive data connections.

Step 15 – Configure the EC2 Public IP

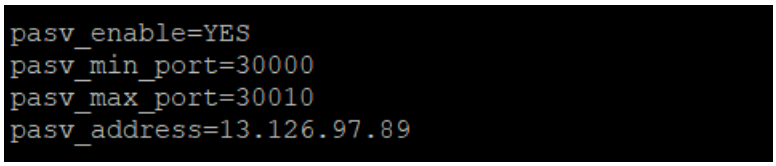
The EC2 public IPv4 address was:

```
13.126.97.89
```

The following setting was added:

```
pasv_address=13.126.97.89
```

This ensures that VSFTPD provides the FTP client with the correct public IP address for passive data connections.



Step 16 – Configure AWS Security Group

The EC2 Security Group was configured to allow:

```
TCP Port 21
Source: My IP
```

The passive FTP port range was also allowed:

```
TCP Ports 30000-30010
Source: My IP
```

Using `My IP` restricts access to the current client IP address instead of allowing FTP connections from anywhere on the internet.

Step 17 – Configure RHEL Firewall

The FTP service was allowed through the RHEL firewall:

```
sudo firewall-cmd --permanent --zone=public --add-service=ftp
```

The passive FTP ports were allowed using:

```
sudo firewall-cmd --permanent --zone=public --add-port=30000-30010/tcp
```

The firewall was reloaded:

```
sudo firewall-cmd --reload
```

The configuration was verified using:

```
sudo firewall-cmd --list-all
```

The output showed the FTP service and passive port range:

```
services: ... ftp ...
ports: 30000-30010/tcp
```

Step 18 – Configure SELinux

The SELinux status was checked:

```
getenforce
```

The result was:

```
Enforcing
```

FTP-related SELinux booleans were checked:

```
sudo getsebool -a | grep ftp
```

The following boolean was available:

```
ftpd_full_access --> off
```

It was enabled permanently using:

```
sudo setsebool -P ftpd_full_access on
```

The setting was verified using:

```
sudo getsebool ftpd_full_access
```

The result was:

```
ftpd_full_access --> on
```

This allows VSFTPD to access the configured FTP directories while SELinux remains in Enforcing mode.

Step 19 – Create a TLS Certificate

To provide secure FTP communication, a TLS certificate and private key were created:

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 -
keyout /etc/pki/tls/private/vsftpd.key -out
/etc/pki/tls/certs/vsftpd.crt
```

The certificate is used to encrypt the FTP login and file-transfer communication.



Step 20 – Configure TLS in VSFTPD

The VSFTPD configuration file was edited:

```
sudo nano /etc/vsftpd/vsftpd.conf
```

The following TLS configuration was added:

```
ssl_enable=YES
allow_anon_ssl=NO
force_local_logins_ssl=YES
force_local_data_ssl=YES

rsa_cert_file=/etc/pki/tls/certs/vsftpd.crt
rsa_private_key_file=/etc/pki/tls/private/vsftpd.key
```

```
ssl_sslv2=NO
ssl_sslv3=NO
```

This enables FTP over TLS and encrypts local-user login credentials and data transfers.

```
ssl_enable=YES
allow_anon_ssl=NO
force_local_logins_ssl=YES
force_local_data_ssl=YES

rsa_cert_file=/etc/pki/tls/certs/vsftpd.crt
rsa_private_key_file=/etc/pki/tls/private/vsftpd.key

ssl_sslv2=NO
ssl_sslv3=NO
```

Step 21 – Restart VSFTPD

The VSFTPD service was restarted to apply the configuration:

```
sudo systemctl restart vsftpd
```

The service was verified using:

```
sudo systemctl status vsftpd
```

The service was confirmed to be active and running.

Step 22 – Verify Secure FTP Using FileZilla

FileZilla was configured with:

```
Protocol: FTP - File Transfer Protocol
Host: 13.126.97.89
Port: 21
Encryption: Require explicit FTP over TLS
Logon Type: Normal
User: zenhal
Password: <configured password>
```

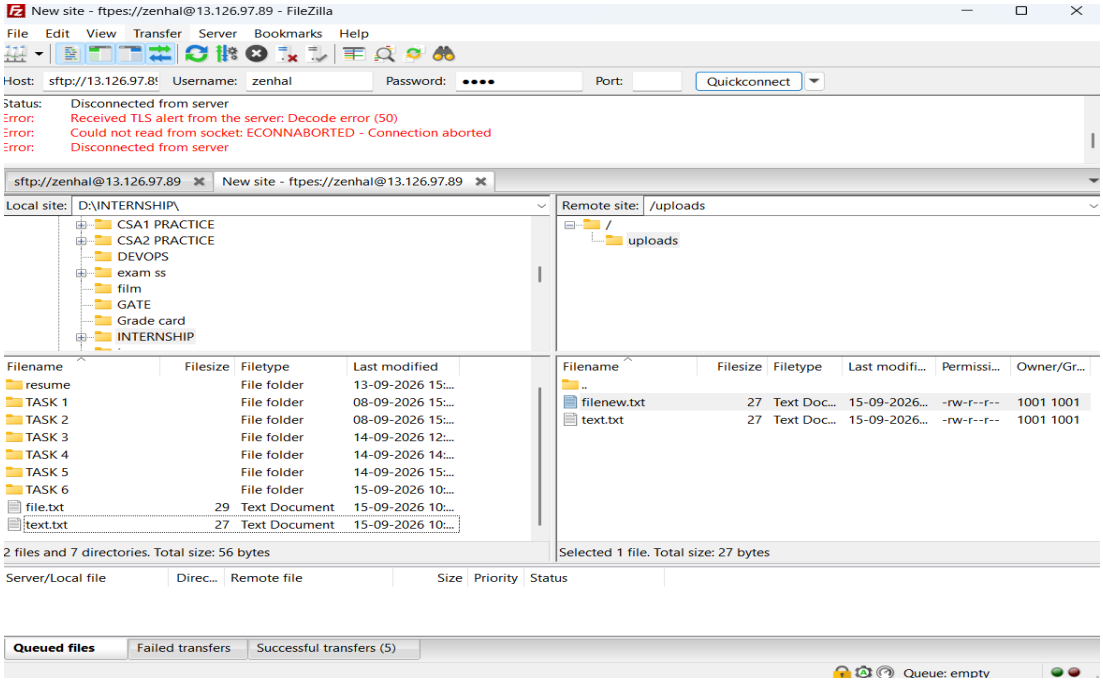
The connection was established using port 21 with explicit TLS encryption.

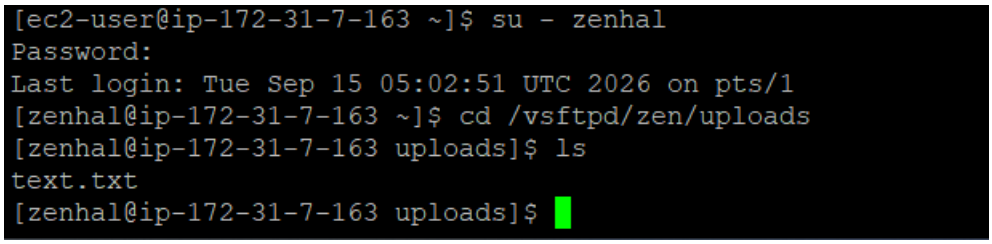
The remote directory provided access to:

```
/
└── uploads
```

A test file was uploaded to the uploads directory to verify the file-transfer operation.

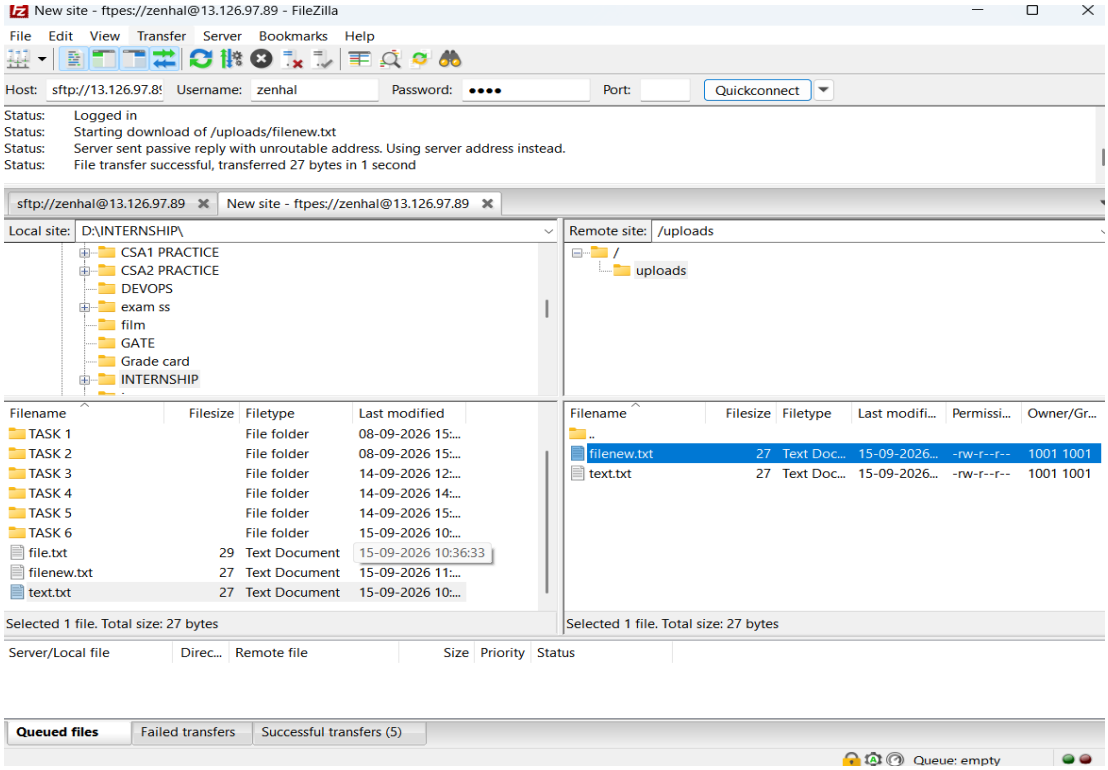
Windows to server:





Server to windows:

```
[zenhal@ip-172-31-7-163 ~]$ cd /vsftpd/zen/uploads
[zenhal@ip-172-31-7-163 uploads]$ ls
text.txt
[zenhal@ip-172-31-7-163 uploads]$ cat >> filenew.txt
hellow this is vsftpd file
^C
[zenhal@ip-172-31-7-163 uploads]$
```



Verification

- SFTP was successfully configured using the SSH service on port 22.
- The `zenhal` user was restricted to SFTP access.
- The SFTP chroot directory was configured as `/sftp/zenhal`.
- The SFTP upload directory was configured as `/sftp/zenhal/upload`.

- SFTP connectivity was successfully verified using FileZilla.
- VSFTPD was successfully installed on the RHEL EC2 instance.
- The VSFTPD service was enabled and confirmed to be active and running.
- The VSFTPD directory was configured as `/vsftpd/zen/uploads`.
- Anonymous FTP access was disabled.
- The `zenhal` user was restricted to the configured FTP directory.
- Passive FTP was enabled using ports 30000–30010.
- The EC2 public IP `13.126.97.89` was configured for passive FTP connections.
- AWS Security Group rules were configured for port 21 and passive FTP ports 30000–30010.
- RHEL firewalld was configured to permit FTP and passive FTP connections.
- SELinux remained in Enforcing mode and `ftpd_full_access` was enabled.
- TLS was configured for VSFTPD to provide encrypted FTP communication.
- Secure FTP was verified using FileZilla through port 21.

Conclusion

Through this task, SFTP and VSFTPD were successfully configured on an AWS EC2 instance running RHEL. SFTP was configured through SSH on port 22 with restricted access for the `zenhal` user. A chroot environment was used to restrict the user's access to the specified SFTP directory.

VSFTPD was installed and configured on port 21 with restricted local-user access. Passive FTP was configured using ports 30000–30010, and the required AWS Security Group and RHEL firewall rules were added. SELinux was maintained in Enforcing mode, and the required FTP access permission was enabled.

TLS encryption was also configured for VSFTPD to provide secure FTP communication. Both SFTP and secure FTP were verified using FileZilla, providing practical experience with restricted file-transfer services, firewall configuration, passive FTP, SELinux, and encrypted file transfer.